Week 6 – Design Studio: Answers

Design the airline's disruption assistant

> Read this after the studio. It isn't the only good answer; it's what a strong answer contains. The system is the retrieval-grounded assistant from the lecture, with a passenger and a flight in place of an employee and a policy. What is different is the rates: sources that change monthly, hourly and by the minute, a question peak in the same hour the documents move fastest, and an answer that commits the airline to money.

The numbers, as given

NUMBER	VALUE	IMPLICATION
questions at the peak	150 a second	the cache carries the load; the vendor, not the index, is the bottleneck
the answer	within 5 s, the waiver quoted	one streamed generation; a money claim is a verbatim quote, so code can check it
fare rules	2,000 documents, monthly	rebuilt into the index on a schedule; nothing about freshness turns on them
waivers	hourly, each with routes, dates and an expiry	updated in the index on the change, in minutes; the expiry rides with the chunk
flight status	1,500 flights, by the minute	never indexed; a table read by flight key at the request
the vendor	50 generations a second	at the peak two questions in three cannot be generated; the cache and the degraded mode are the design, not an extra
unanswerable	about a third	the floor and the handoff are a third of the work, not an edge case

1. Architecture

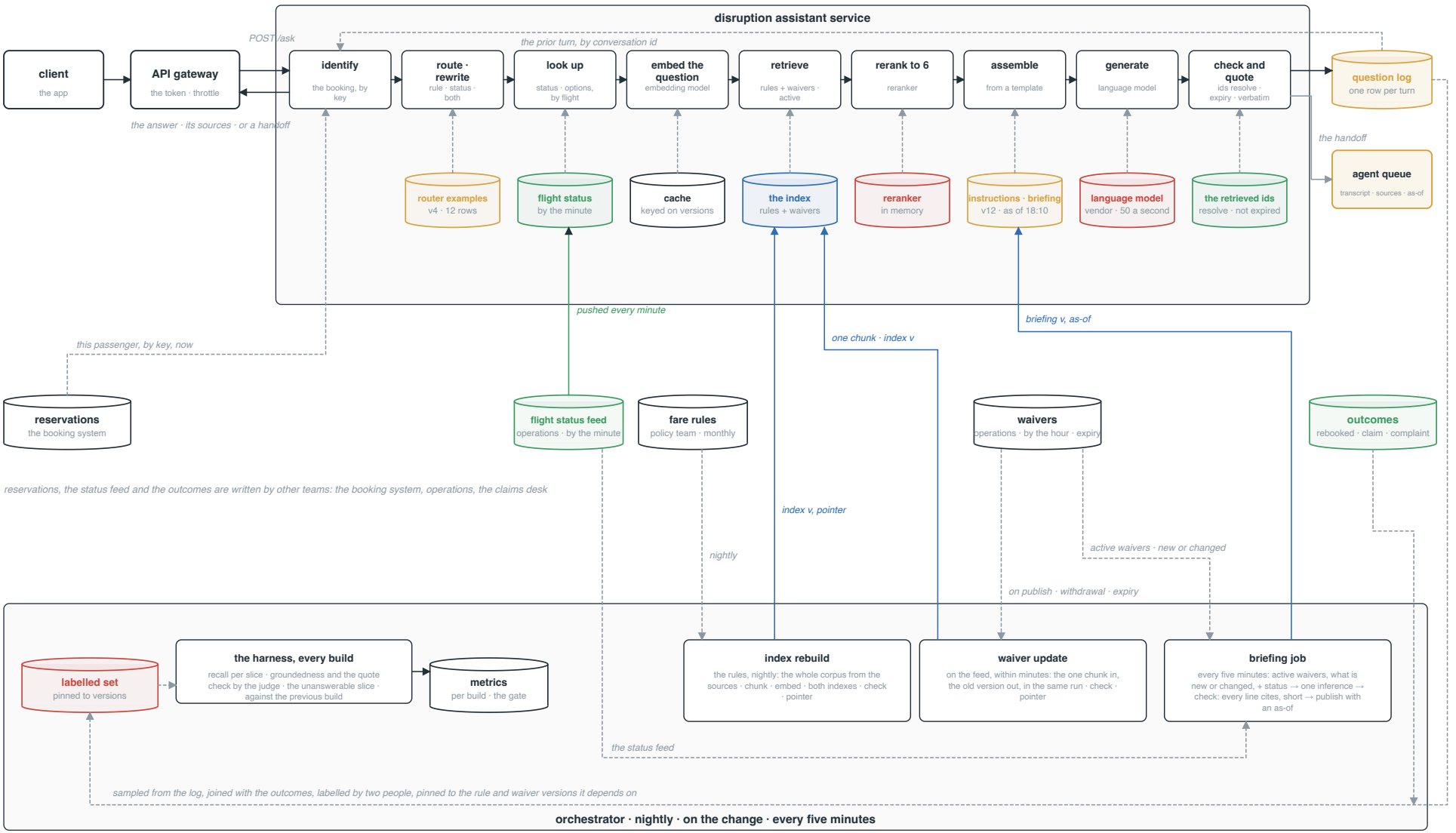
One mechanism per rate

Each source reaches an answer by the mechanism its rate demands, and the mechanisms stay separate.

SOURCE	MECHANISM	HOW OLD AT THE ANSWER
fare rules	the index rebuild, nightly: every rule at its current version, chunked by clause with the heading path, embedded with the pinned model, into both indexes; the check; the pointer	up to a day
waivers	the update on the change: the waiver feed fires on publish, extension or withdrawal; the one chunk embedded and inserted, the superseded chunk deleted in the same run; the check; the pointer	minutes, measured as publish-to-pointer and withdrawal-to-pointer
flight status	a function call at the request: operations overwrites a table every minute; read by flight key; the minute read goes into the context and the log	a minute
booking	a read by key from the reservation system at the identify stage; never embedded, because it changes per request and the index is shared	now
the briefing	a job every five minutes: one inference over the active waivers and the status feed, with its own instructions; code checks that every line cites a waiver id or the feed and that the whole is short; published with a version and an as-of, or the previous one stays	five minutes; its as-of is in every answer

The waivers are in the index as rules and in the briefing as news. A chunk says what W-2041 allows; the briefing says W-2041 was extended at 20:10 and is new since the last run.

The request path



The disruption assistant: the service, the sources at their rates, the orchestrator, and the flow between them. Solid: the request path and writes. Dotted: reads. Blue: the index and the briefing, versioned.

STAGE	WHAT IT DOES	CODE OR MODEL
identify	the passenger from the token; the booking by key; the prior turn by conversation id	reads; no model
route · rewrite	RULE, STATUS or BOTH, from a dozen worked examples in the prompt; a follow-up rewritten to stand alone	one inference, parsed by code
look up	status, gate, new time and rebooking options, from the status table by flight key	a table read; the minute recorded
embed	the rewritten question	a model; skipped on a cache hit
retrieve	lexical and vector candidates, filtered inside the search to rules and to waivers active on this route and these dates; fused	the index, versioned
rerank to 6	the question against each candidate	a model, pinned
assemble	from a template: the instructions, the briefing with its as-of, the lookups with their minute, six passages with ids and expiries, the prior turn, the question	code, against a budget
generate	one streamed inference; every fact cited as [n]; any sentence that commits money quoted verbatim with the waiver id	the language model, pinned
check and quote	every [n] is a retrieved id; every quote is a substring of its passage; every quoted waiver's status and expiry re-read from the waivers table now; not NOT_FOUND; within length. A failing draft is never shown	code, on every answer; the judge samples behind it

The seam. The index, the briefing, the status table and the router's examples each carry a version or an as-of. The request path reads them and never writes them.

The cache

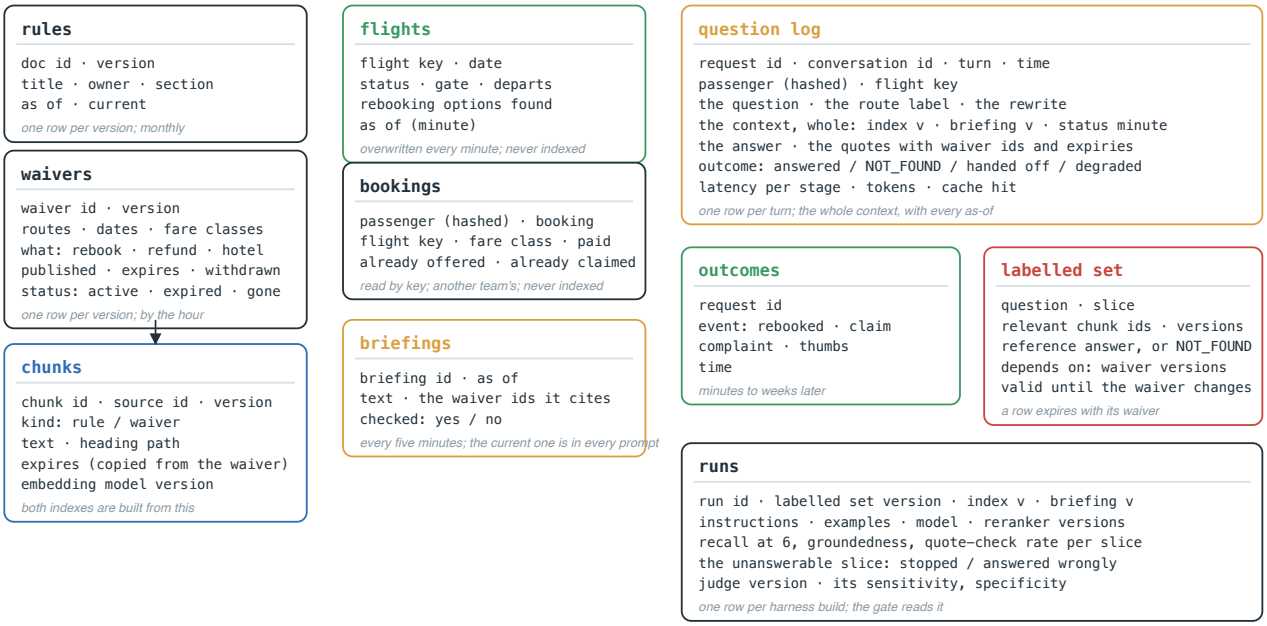
The key is the rewritten question, normalized, with the index version, the briefing version and the status minute for this flight key. The value is the finished answer with its quotes and ids. Anything that could make an answer wrong moves part of the key, so a stale entry is simply never looked up again; nothing is invalidated by hand. The booking stays outside the key: the answer is written for the flight, and the identify stage adds the passenger's own lines around the cached text. The hit rate in the storm is what makes 150 a second possible on 50 generations.

The dials

How often the briefing runs; how old a status may be before the look-up refuses it; how many candidates each index returns and how many the reranker keeps; the floor; the context budget. Each trades

freshness or recall for cost, and each is measured before it is set.

2. Data handling



The tables of the disruption assistant. Flights and bookings are read at the request and never indexed; a labelled question expires with the waiver it depends on.

Rules and waivers are kept by version, and a version is never edited. Chunks carry the source id, the version, the kind and, for a waiver, the expiry copied from the waiver, so the filter and the check read it without a join. Flights and bookings are other teams' tables, read by key and never indexed. The question log is one row per turn and holds the whole context with every as-of. Outcomes join on the request id, minutes to weeks later.

A waiver as a document

COLUMN	VALUE
waiver id · version	W-2041 · v2
routes · dates · fare classes	any flight through the hub · today and tomorrow · all
what	rebook once without fee within seven days; a hotel tonight if the onward flight is tomorrow; no cash refund
published · expires · withdrawn	18:05 · 23:59 tomorrow · (empty)
status	active

Its chunk is the waiver's text, whole, headed "Waiver W-2041 v2 · hub closure", with the routes and dates in the text so the lexical index finds them and `expires` in the metadata. When v3 is published at 20:10, v2's status becomes superseded, v3 is a new row, and the update inserts v3's chunk and deletes v2's in one run. The minute a waiver expires or is withdrawn, the same job removes its chunk. Until that run lands, the retrieval filter excludes the chunk by its expiry and the check re-reads its status, so no answer quotes it.

The labelled set, pinned

A row holds the question, its slice (rule, status, both, unanswerable; and rebooking, refund, hotel, baggage), the relevant chunk ids at their versions, the reference answer or NOT_FOUND, and the waiver versions it depends on. Two labellers, their agreement with its interval, adjudication, the rule amended. The harness runs a row against an index built from the versions on the row, not tonight's. A withdrawn waiver does not make its rows wrong; it makes them a replay, and a new row is labelled for the new state. Some rows of the unanswerable slice are last week's answerable rows whose waiver has gone.

3. Measurement

Before any passenger sees it

STAGE	MEASURED	AGAINST
the router	accuracy per label; BOTH routed as RULE is the failure that matters	the labelled slice, three runs
retrieval	recall at 6 per retriever and per slice; the waiver slice on its own, because route codes are found by the lexical index	the relevant chunk ids, at their versions
the reranker	recall at 6 after reranking	the same
the floor	of the unanswerable slice, how many stopped; of the answerable, how many stopped wrongly	both slices
generation	groundedness per claim and correctness, by the judge; the quote check's pass rate, in code	the judge validated against the team's labels: sensitivity, specificity, kappa, the corrected rate
the briefing	groundedness of every line against what it cites	its own labelled sample
the check	rejections by reason: bad citation, quote not a substring, waiver expired or withdrawn	every answer; no labels needed
the whole	the report row per slice	the previous build

The unanswerable third. Compensation the rules do not give, a promise an agent made by phone, a hotel the airline does not offer tonight, and questions answerable last week under a waiver that has gone. The floor stops the ones whose best reranker score is under the threshold. The model answers the rest anyway, and the measure is how often it says NOT_FOUND rather than writing fluently from a passage that does not apply.

With passengers

The primary metric is a rebooking completed in the session after an answer that offered one, per slice, against the previous version in an interleaved or split rollout, with its interval. The guardrail that stops a release whose primary rose: the quote check's rejection rate and the judge's groundedness on the money slices, refund and hotel. A version that lifts rebookings by writing more confident answers shows up here first. A second guardrail is the count of answers whose quoted waiver was withdrawn before the answer was shown, measured on every answer and held near zero.

The pipeline's own numbers

Publish-to-pointer and withdrawal-to-pointer per waiver. The briefing's age at each answer. The status table's age per flight. The cache hit rate and the age of the entry served. The vendor's queue depth and the share of answers served degraded. Each is a column in the log row, so each is a chart.

Bonus: the degraded mode

At 150 a second against 50 generations, two questions in three cannot be generated. First, the cache: in a storm most questions are the same question. Second, on a cache miss when the vendor's queue is over budget, the assistant skips generation and shows the search result: the flight's status and options from the table, and the active waiver quoted whole with its id and expiry, under a heading that says no summary could be written. Every fact shown is still a verbatim quote or a status read. The log row records `degraded`. What the system never does is widen the budget by skipping the check, cutting the passages below the waiver's condition, or answering from the briefing alone. A degraded answer is a smaller true answer, never a faster guess.

The hotel at 19:40

What the log holds. The row for that turn: the passenger, hashed; the flight key; the question and its rewrite; the index version and its as-of; the briefing version and its as-of; the status minute; the six passages with ids, versions and expiries, one of them the hotel clause; the context, whole; the draft with its quote and waiver id; the check's result; the answer shown. Everything needed to say what the system knew at 19:40.

Which component let it through. The waiver update. The withdrawal at 19:30 should have started it, and the pointer had not moved by 19:40. Ten minutes against a freshness number of minutes means the update failed its check, the feed was late, or the withdrawal was left for the nightly rebuild, which is the one design this answer rules out. The withdrawal-to-pointer line on the dashboard says which. Retrieval did what it should with the index it had; generation quoted a passage it was given.

Which component should have stopped it. The check. Its last rule reads the quoted waiver's status and expiry at the moment of the answer, from the waivers table, not from the chunk's copy. A quote from a withdrawn version fails the check, and the passenger sees the floor, not the promise. Without that rule, freshness is only as good as the slowest build. With it, a second guarantee holds while the first catches up, and the window between them is measured.